2

Restricting and Sorting Data

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Objectives

After completing this lesson, you should be able to do the following:

- Limit the rows retrieved by a query
- Sort the rows retrieved by a query

Limiting Rows Using a Selection

EMP

EMPNO	ENAME	JOB	...	DEPTNO
7839	KING	PRESIDENT		10
7698	BLAKE	MANAGER		30
7782	CLARK	MANAGER		10
7566	JONES	MANAGER		20
...				

"…retrieve all employees in department 10"

EMP

EMPNO	ENAME	JOB	...	DEPTNO
7839	KING	PRESIDENT		10
7782	CLARK	MANAGER		10
7934	MILLER	CLERK		10

Limiting Rows Selected

- Restrict the rows returned by using the WHERE clause.

```
SELECT        [DISTINCT] {*| column [alias], ...}
FROM          table
[WHERE        condition(s)];
```

- The WHERE clause follows the FROM clause.

Using the WHERE Clause

```
SQL> SELECT ename, job, deptno
  2  FROM   emp
  3  WHERE  job='CLERK';
```

```
ENAME      JOB       DEPTNO
---------- --------- ---------
JAMES      CLERK            30
SMITH      CLERK            20
ADAMS      CLERK            20
MILLER     CLERK            10
```

Character Strings and Dates

- Character strings and date values are enclosed in single quotation marks.
- Character values are case sensitive and date values are format sensitive.
- The default date format is DD-MON-YY.

```
SQL> SELECT   ename, job, deptno
  2  FROM     emp
  3  WHERE    ename = 'JAMES';
```

Comparison Operators

Operator	Meaning
=	Equal to
>	Greater than
>=	Greater than or equal to
<	Less than
<=	Less than or equal to
<>	Not equal to

Using the Comparison Operators

```
SQL> SELECT ename, sal, comm
  2  FROM   emp
  3  WHERE  sal<=comm;
```

```
ENAME             SAL      COMM
---------- ---------- ---------
MARTIN           1250      1400
```

Other Comparison Operators

Operator	Meaning
BETWEEN ...AND...	Between two values (inclusive)
IN(list)	Match any of a list of values
LIKE	Match a character pattern
IS NULL	Is a null value

Using the BETWEEN Operator

Use the BETWEEN operator to display rows based on a range of values.

```
SQL> SELECT   ename, sal
  2  FROM     emp
  3  WHERE    sal BETWEEN 1000 AND 1500;
```

```
ENAME            SAL
---------- ---------
MARTIN          1250
TURNER          1500
WARD            1250
ADAMS           1100
MILLER          1300
```

Lower limit

Higher limit

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Using the IN Operator

Use the IN operator to test for values in a list.

```
SQL> SELECT   empno, ename, sal, mgr
  2  FROM     emp
  3  WHERE    mgr IN (7902, 7566, 7788);
```

```
    EMPNO ENAME             SAL       MGR
--------- ---------- ---------- ---------
     7902 FORD             3000      7566
     7369 SMITH             800      7902
     7788 SCOTT            3000      7566
     7876 ADAMS            1100      7788
```

Using the LIKE Operator

- Use the LIKE operator to perform wildcard searches of valid search string values.
- Search conditions can contain either literal characters or numbers.
  - % denotes zero or many characters.
  - _ denotes one character.

```
SQL> SELECT   ename
  2  FROM     emp
  3  WHERE    ename LIKE 'S%';
```

Using the LIKE Operator

- You can combine pattern-matching characters.

```
SQL> SELECT   ename
  2  FROM     emp
  3  WHERE    ename LIKE '_A%';
```

```
ENAME
----------
MARTIN
JAMES
WARD
```

- You can use the ESCAPE identifier to search for "%" or "_".

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Using the IS NULL Operator

Test for null values with the IS NULL operator.

```
SQL> SELECT  ename, mgr
  2  FROM    emp
  3  WHERE   mgr IS NULL;
```

```
ENAME            MGR
---------- ---------
KING
```

Logical Operators

Operator	Meaning
AND	Returns TRUE if *both* component conditions are TRUE
OR	Returns TRUE if *either* component condition is TRUE
NOT	Returns TRUE if the following condition is FALSE

Using the AND Operator

AND requires both conditions to be TRUE.

```
SQL> SELECT empno, ename, job, sal
  2  FROM   emp
  3  WHERE  sal>=1100
  4  AND    job='CLERK';
```

```
    EMPNO ENAME      JOB             SAL
--------- ---------- --------- ---------
     7876 ADAMS      CLERK          1100
     7934 MILLER     CLERK          1300
```

Using the OR Operator

OR requires either condition to be TRUE.

```
SQL> SELECT empno, ename, job, sal
  2  FROM   emp
  3  WHERE  sal>=1100
  4  OR     job='CLERK';
```

```
    EMPNO ENAME      JOB             SAL
--------- ---------- --------- ---------
     7839 KING       PRESIDENT      5000
     7698 BLAKE      MANAGER        2850
     7782 CLARK      MANAGER        2450
     7566 JONES      MANAGER        2975
     7654 MARTIN     SALESMAN       1250
     ...
     7900 JAMES      CLERK           950
     ...
14 rows selected.
```


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Using the NOT Operator

```
SQL> SELECT ename, job
  2  FROM   emp
  3  WHERE  job NOT IN ('CLERK','MANAGER','ANALYST');
```

```
ENAME      JOB
---------- ---------
KING       PRESIDENT
MARTIN     SALESMAN
ALLEN      SALESMAN
TURNER     SALESMAN
WARD       SALESMAN
```

Rules of Precedence

Order Evaluated	Operator
1	All comparison operators
2	NOT
3	AND
4	OR

Override rules of precedence by using parentheses.

Rules of Precedence

```
SQL> SELECT ename, job, sal
  2  FROM   emp
  3  WHERE  job='SALESMAN'
  4  OR     job='PRESIDENT'
  5  AND    sal>1500;
```

```
ENAME      JOB             SAL
---------- --------- ---------
KING       PRESIDENT      5000
MARTIN     SALESMAN       1250
ALLEN      SALESMAN       1600
TURNER     SALESMAN       1500
WARD       SALESMAN       1250
```

Rules of Precedence

Use parentheses to force priority.

```
SQL> SELECT    ename, job, sal
  2  FROM      emp
  3  WHERE     (job='SALESMAN'
  4  OR        job='PRESIDENT')
  5  AND       sal>1500;
```

```
ENAME      JOB             SAL
---------- --------- ---------
KING       PRESIDENT      5000
ALLEN      SALESMAN       1600
```


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ORDER BY Clause

- Sort rows with the ORDER BY clause
  - ASC: ascending order, default
  - DESC: descending order
- The ORDER BY clause comes last in the SELECT statement.

```
SQL> SELECT    ename, job, deptno, hiredate
  2  FROM      emp
  3  ORDER BY hiredate;
```

```
ENAME      JOB           DEPTNO HIREDATE
---------- --------- ---------- ---------
SMITH      CLERK             20 17-DEC-80
ALLEN      SALESMAN          30 20-FEB-81
...
14 rows selected.
```


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Sorting in Descending Order

```
SQL> SELECT    ename, job, deptno, hiredate
  2  FROM      emp
  3  ORDER BY hiredate DESC;
```

```
ENAME      JOB          DEPTNO HIREDATE
---------- --------- --------- ---------
ADAMS      CLERK            20 12-JAN-83
SCOTT      ANALYST          20 09-DEC-82
MILLER     CLERK            10 23-JAN-82
JAMES      CLERK            30 03-DEC-81
FORD       ANALYST          20 03-DEC-81
KING       PRESIDENT        10 17-NOV-81
MARTIN     SALESMAN         30 28-SEP-81
...
14 rows selected.
```


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Sorting by Column Alias

```
SQL> SELECT    empno, ename, sal*12 annsal
  2  FROM      emp
  3  ORDER BY annsal;
```

```
    EMPNO ENAME         ANNSAL
--------- ---------- ---------
     7369 SMITH           9600
     7900 JAMES          11400
     7876 ADAMS          13200
     7654 MARTIN         15000
     7521 WARD           15000
     7934 MILLER         15600
     7844 TURNER         18000
...
14 rows selected.
```

Sorting by Multiple Columns

- The order of ORDER BY list is the order of sort.

```
SQL> SELECT    ename, deptno, sal
  2  FROM      emp
  3  ORDER BY  deptno, sal DESC;
```

```
ENAME         DEPTNO       SAL
---------- --------- ---------
KING              10      5000
CLARK             10      2450
MILLER            10      1300
FORD              20      3000
...
14 rows selected.
```

- You can sort by a column that is not in the SELECT list.

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Summary

```
SELECT      [DISTINCT] {*| column [alias], ...}
FROM        table
[WHERE      condition(s)]
[ORDER BY   {column, expr, alias} [ASC|DESC]];
```

Practice Overview

- Selecting data and changing the order of rows displayed
- Restricting rows by using the WHERE clause
- Using the double quotation marks in column aliases